

New technology for reducing operational costs

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A wide range of innovative techniques are currently being developed to tackle access network operations and maintenance. These include both addressing the operational requirements of new technology and driving up the profitability and quality of service associated with existing networks and services. A number of such techniques are discussed, all focused on delivering improved quality of service to customers at a similar, or lower, cost than current solutions achieve. The sheer scale of the access network, which can account for the majority of a telco's cost base, makes the potential benefits from these techniques enormous. An appreciation of the diversity, innovation and business benefit being achieved from current developments in the field of access operations and maintenance is given.

1. Introduction

New technology in the access network is aimed mostly at enabling fresh services to be delivered to customers. New technology can also be used to reduce the cost of operations, administration and maintenance (OAM). There are two primary business drivers for this:

- achieving the right quality of service at lower cost,
- ensuring that current account costs associated with new network technology are competitive.

There are a number of OAM technology solutions currently being developed or researched that will make a significant impact on access network operations and maintenance costs and hence support these business drivers. An overview of these solutions is given to convey the diversity, innovation and business benefit currently being achieved.

The main factors that determine current account costs and quality of service are those associated with the following business processes:

- providing service to a new customer,
- repairing the network,
- routine maintenance of the network.

The solutions described are all aimed at improving the effectiveness of one or more of these processes. They fall into three main themes, as illustrated in Fig 1.

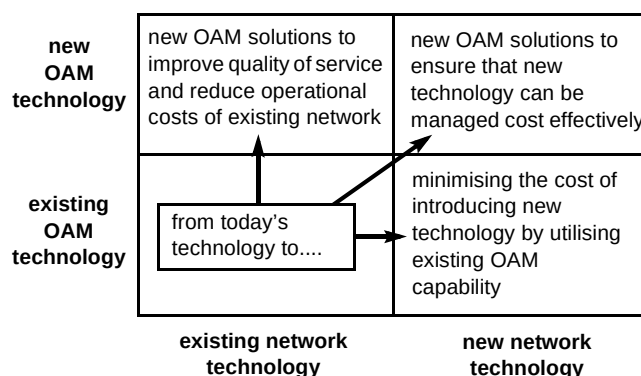


Fig 1 Developing OAM solutions for both new and existing technology.

2. New OAM technology for existing networks

There is little doubt that future revenue streams will depend more and more on novel services that use new transmission technology. However, for the immediate future, profitability will be determined primarily by existing services, most notably PSTN based. For this reason, there is still a significant requirement to develop improved methods for the OAM of the existing network technology — which for the access network means basic, often ageing, twisted pairs with little or no active electronics.

2.1 Measuring network health

Overnight routine testing of the access network using remote line test systems has been commonplace in some telcos for many years. Up until now, making use of the routine test information has been difficult, as effective work prioritisation algorithms have not been available. New

technology has recently been developed to address this issue. Sophisticated processing algorithms are used to produce a measure of the electrical 'health' of each distribution point (DP¹) within the network. The principal objective of these algorithms is to ensure that any work undertaken on routine maintenance of the network will be targeted at those parts of the network most likely to fail, and hence deliver maximum benefit.

The key components are shown in Fig 2. A database of routine line test results is built. This information is processed, along with information about network connectivity extracted from electronic network records, using proprietary algorithms developed within BT. The result of this processing is a list of DPs with associated 'health' scores. The score gives, for each DP, the probability that one or more of the pairs routed through the DP will fail within three months.

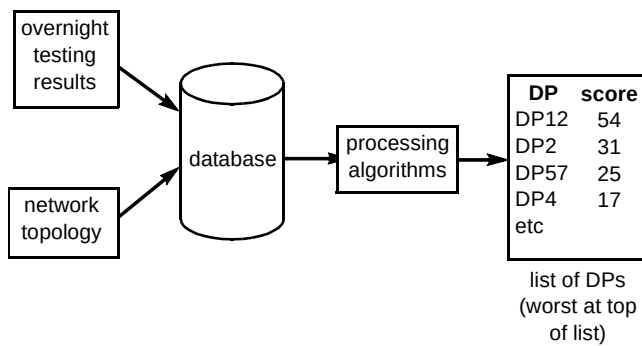


Fig 2 Using routine line test results to determine network health.

The processing algorithms have several steps as illustrated in Fig 3. The first task is to remove suspect measurement results from the analysis. These may appear due to problems with the line test system test head or the presence of unexpected voltage on the line under test. The next step is to look at information about the line held on an electronic plant record database to identify those lines that might have unusual electrical conditions on them which could disrupt the health scoring process. Examples here are

field engineer's lines which terminate as an open circuit at the cabinet and lines routed through pair-gain equipment.

The remaining data is then tested against a set of rules which identify 'hard' fault conditions. These are conditions such as a total break in a leg of the pair or a serious short circuit. The leakage measurements² for each pair to a given DP are then processed to produce an overall score that describes the electrical condition of the pairs routed to that DP. This is repeated until a health score is available for all DPs. The final step is to produce a prioritised list of work taking into account both hard faults and DP health scores.

These algorithms have been tested in large-scale pilot schemes within BT. A significant reduction in customer fault report volumes was observed, as well as a reduction in the susceptibility to damage of the pilot areas during adverse weather conditions.

2.2 Where is the fault?

Having identified that there is a network fault, either from an anticipation system such as that described in section 2.1, or from a customer fault report, the challenge is to identify the location of the problem so that an engineer can be despatched to effect a repair.

Current line test systems aim to locate the fault to one of three sections of the network:

- customer's premises and dropwire,
- the local network,
- the exchange.

¹ The distribution point is the last flexibility point in the network before the customer. Each DP typically serves between 10 and 20 customers, and there is a dedicated connection between the DP and the customer premises. DPs are typically located at the top of poles or in underground enclosures.

² 'Leakage' is the impedance between a given leg of a twisted pair and either exchange earth, exchange battery or the other leg of the pair. Ideally this should be close to infinite. Low values of impedance are usually caused by a breakdown in the insulation of the wire or moisture giving rise to an electrical path between exposed conductors at a joint in the cable.

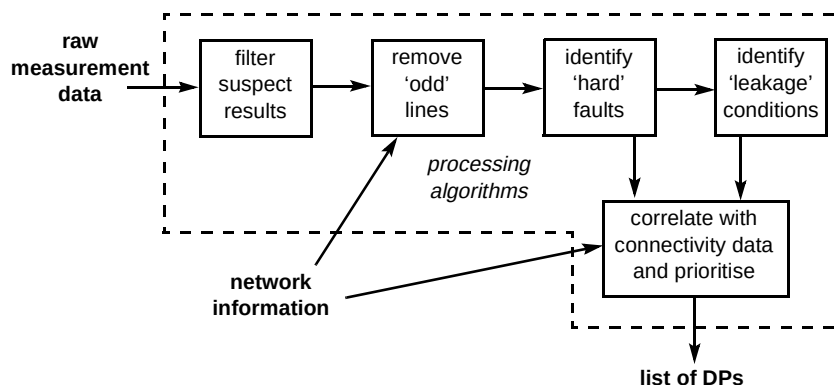


Fig 3 Network health estimation — processing algorithms.

There is a significant opportunity for reducing operational costs if faults in the local network can be localised with greater precision, for example, to a specific network node such as a cabinet, DP or joint box. The cost savings arise from reducing the travelling and investigatory time required by the repair engineer.

Currently, the only distance estimate available from line test systems is the distance to fault from the exchange. These are not generally used for two reasons:

- it is too difficult to translate from the distance estimate to a geographical location,
- the estimates are unreliable in many cases as the effect of the fault on the distance calculation is not considered.

BT now has a prototype system that uses more sophisticated algorithms to process the line test data and produce an estimate of the fault's distance from the customer's premises.

Estimating distance from customer's premises rather than the exchange gives some distinct advantages, as most network faults occur much closer to the customer's premises than the exchange:

- the distance estimates are generally much shorter, and hence the potential error in the estimate is reduced — the errors generally accrue from unknown factors in converting from electrical parameters to distance, such as cable type and unknown or unexpected cable routings,
- it becomes much simpler to convert from distance estimate to geographical location using paper-based plant records, as only a small section of the plant record needs to be consulted,
- when the distance from customer estimate is short (say a few tens of metres), there is no need to consult the plant records at all and considerable time and effort is saved.

The algorithm used to compute the 'distance from customer' estimate is illustrated in Fig 4.

It is the ability to test routinely almost all lines that provides the enabler for calculating distance from the customer's premises. The algorithm takes the results for a given day and determines, for each pair tested, whether or not the result is valid. If it is, then it is used to compute a new rolling average figure for the capacitance between a leg

of the pair and earth. This rolling average represents the capacitance for the line in a working state.

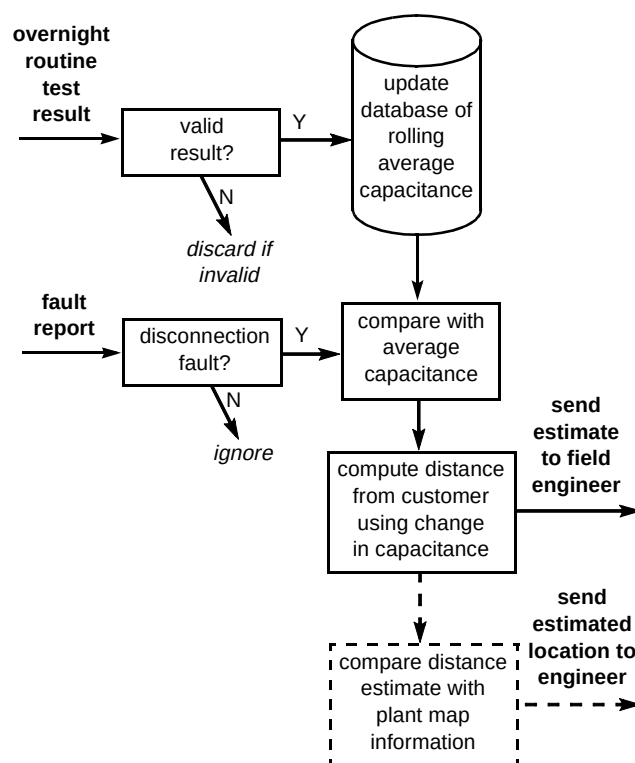


Fig 4 Distance to fault from customer algorithm.

When a fault report is received, the first step in the algorithm is to determine whether the fault is caused by a disconnection. If it is, then the test results of the line after it has been reported faulty are compared with the stored rolling average. The difference in the two capacitance values can then be used to produce an estimate of the distance of the fault from the customer's premises. This is passed to the field engineer and assists with the location of the problem.

Additional value can be obtained by comparing the distance estimate with the distance to key network elements recorded on plant records. This could, for example, identify that the fault is at a cross-connection point and enable the system to give precise network node location information to the field engineer. Most faults occur at network nodes so this has a significant impact on repair times.

An additional benefit arising from this distance estimation algorithm is the ability to identify a common fault cause. If several faulty customer lines have a similar distance estimate, it is likely that there is one problem underlying them and a single engineer can be assigned to their repair.

2.3 Using an active line termination

Much of the technology described in this paper is aimed at improving the interpretation of test results taken using line test equipment situated in the exchange. The value of

these measurements can be significantly enhanced if it is possible to control electrical conditions at the remote end of the line.

An intelligent network terminator for twisted pair lines has been trialled which provides such a capability. The prototype terminator is a compact and inexpensive electronic unit that is housed within the customer's network termination. The active part of the terminator is remotely controlled by the line test system as shown in Fig 5. A number of benefits arise from this technology.

- **Enhanced fault demarcation**

Wiring within the customer premises is not usually the telco's responsibility to repair. It can be very difficult to determine from line test results that a fault near the end of the line is in the customer wiring and not the telco's network. This uncertainty results in a large number of unnecessary customer visits. The intelligent terminator can isolate the customer wiring and equipment from the line, under the control of the line test system, allowing only the telco's wiring to be tested. Comparing the line test results in the isolated and normal conditions allows the position of the fault to be determined with a high degree of confidence.

- **Insertion loss measurement**

Loop insertion loss is a good predictor of a loop's ability to support digital subscriber line (DSL) technologies (see also section 3.1). The intelligent terminator includes a wideband signal generator that can be activated by the line test system. An enhanced line test system can measure the signal received at the exchange end of the line yielding the insertion loss at any frequency of interest. This technology could be integrated into the pre-qualification process (described later in section 3.1, Fig 9) and would deliver an extremely accurate means of making the initial

decision, potentially reducing the number of 'unknown' decisions to zero and hence providing a fully one-touch solution.

- **Loop resistance measurement**

The prototype terminator enables direct measurement of the loop resistance by connecting a 600 Ω resistor across the wires of the pair at the customer's premises. This information allows some faults to be detected, such as series resistance faults, which cannot be done using existing line test techniques.

A large trial of the intelligent network terminator technology, using a prototype line test system, has been completed. 2400 lines were equipped with the new device instead of the normal line terminator. Over a period of a year all faults on the trial lines were analysed using the new capabilities in addition to the business as usual methods.

The trial has indicated that the device can significantly improve the accuracy of line tests for certain fault conditions:

- the device enables the confident identification of customer wiring faults — no faults that were diagnosed as customer wiring subsequently turned out to be network faults,
- insertion loss of the line can be measured accurately and quickly giving a clear indication of lines that will not support DSL transmission systems,
- the loop resistance of a large number of lines has been measured every day for almost a year, allowing the variation of loop resistance with time to be measured — this is a vital input to the planning processes for DSL technology.

The trial technology has proved to be a success and work is now under way to refine the design of the

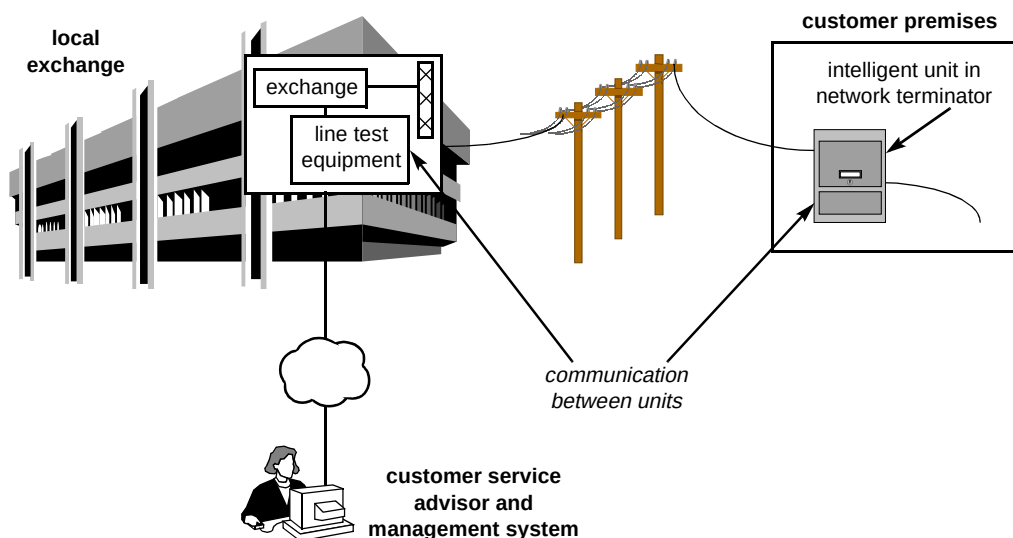


Fig 5 Active twisted pair line termination device.

terminator and to determine the most cost-effective way of deploying it in the network.

2.4 Noise location

Noise faults are becoming an increasing issue for telecommunications companies. Not only have customer expectations risen, but widespread use of voice-band modems means noise problems can prevent, rather than just degrade, service from the customer's point of view. Furthermore, modem technology can in some cases give customers a quantitative measure of noise levels present on their connection.

The technology for dealing with noise problems has fallen behind that available for other network faults. The key problem is that noise faults are almost always intermittent, so that when a measurement of the line is taken the fault condition that causes the noise may not be present and the line may appear electrically sound. Indeed, some causes of noise may not be due to an electrical defect in the line, but to interference from external sources such as railway locomotive power feeds or electric fences. These noise problems frequently cause repeat fault reports with associated poor customer satisfaction and wasted engineering time.

To tackle the problem of diagnosing and curing the more problematic noise problems, a prototype noise recording system has been developed. This system enables customers to record, as soon as noise is heard, 30-second portions of any telephone call that they have initiated by pressing a key on their keypad. The recording is subsequently listened to by a diagnostic engineer and, more often than not, the engineer will be able to suggest a likely cause from audible characteristics of the noise.

This system exploits intelligent network (IN) technology. It is built as an application running on one of the existing IN platforms and hence provides a centralised recording capability. This approach has two key features that make it cost effective to deploy:

- the cost of the application and associated hardware can be spread across a very large number of customers as it can be accessed over the PSTN,
- no field engineering work is required to deploy the system.

The basic operation of the system is illustrated in Fig 6. The customer who is experiencing noise problems wishes to make a call. Rather than dial the required number directly, they will dial a freefone number which connects them to the IN platform. The IN platform then returns a second dial tone to the customer who then dials the number of the person they wish to call and is connected in the usual way. At any time during this call, if noise is heard, the customer can press a number on the telephone dial and create a 30-second recording which will include 15 seconds prior to the

keypress and 15 seconds after the keypress. The recording is automatically stored on a secure file server which can be accessed by a diagnostic engineer from any remote location.

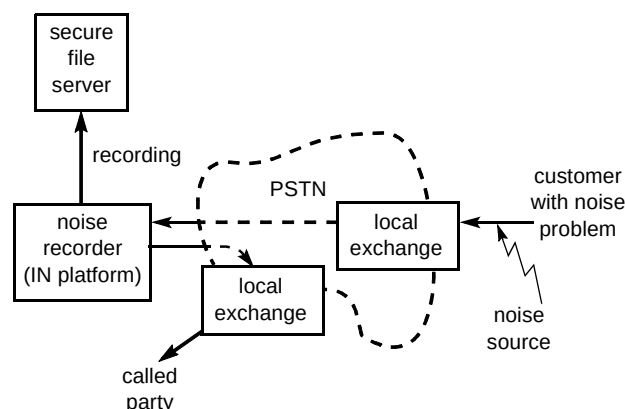


Fig 6 Basic operation of the customer-controlled noise recorder.

If the value of these recordings is found to be high, the next step will be research into algorithms to automate the interpretation of the recorded information.

2.5 Monitoring pressurised cable condition

The final example of using new OAM technology to support existing network technology is a system for monitoring cable pressure in the large access network cables that are, typically, used between the local exchange and the cabinet.

It has long been accepted that pressurisation of these cables has significant benefits in terms of preventing failures due to water ingress into the cable. The new system allows the pressure to be remotely monitored at various points along the cable to ensure that adequate pressure is being maintained. The basic system architecture is shown in Fig 7.

Low-cost pressure transducers are placed at strategic points along the pressurised cable network. A simple communications protocol is used to communicate between data collection equipment (DCE) in the local exchange and each individual transducer. The DCE is also able to monitor the flow of air into the pressurised cables and the duty cycle of the compressor that provides the air.

This information is collected automatically by the DCE on a regular basis. A remote computer system dials in to the DCE periodically, uploads the information held there, and analyses it to identify cables that appear to be deteriorating (usually due to a breakdown in the cable sheath material at a joint).

A key component in the system is the prioritisation algorithm. This is able to identify those cables which, if not attended to, are most likely to suffer from customer service-affecting faults. This enables the most cost-effective

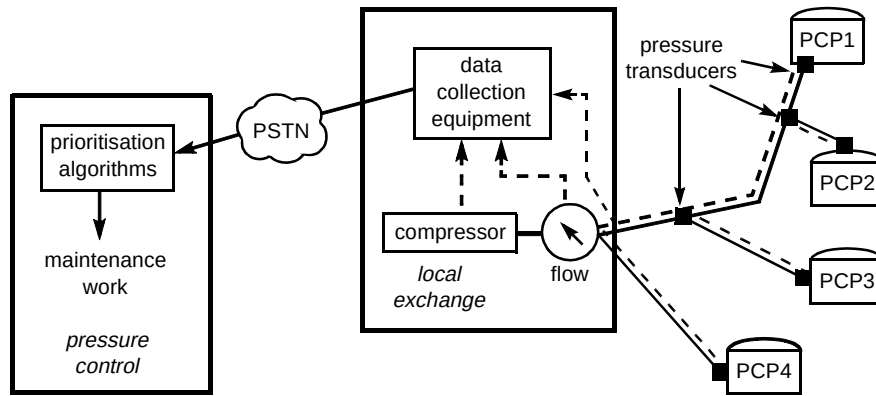


Fig 7 Remote pressurisation monitoring system.

maintenance work to be actioned first, thus making best use of available engineering resources.

An additional benefit of the system is the ability to raise an alarm within minutes of a serious network problem being detected — for example, a sudden loss of pressure due to a cable dig-up or a failure of the compressor unit.

The low cost per line of this system, a key design requirement from the outset, means that full deployment to cover all pressurised cables in the network should be possible.

Future enhancements to the system, using the same low-cost monitoring and data collection technology, could include humidity monitoring to help to identify cables most at risk of water damage, and light level monitoring which would enable joints that are no longer sealed against atmosphere to be detected.

Both of these enhancements could apply to the entire access network rather than just the pressurised cables. They could also deliver a cost-effective solution for the management of fibre cables.

3. Using existing OAM technology to manage future networks

The next generation of network technology that will be introduced into access networks will depend heavily on digital subscriber line systems. This enables delivery of many broadband services over existing twisted-pair networks.

Existing twisted-pair network management capabilities can be applied to DSL systems to reduce both the operational and development costs associated with the technology. Two examples are given here.

3.1 Line test for loop qualification

The ability to determine whether or not a particular customer's existing twisted-pair line is capable of supporting DSL technologies is a key enabler for the deployment of broadband services to a mass-market. This is sometimes referred to as 'loop pre-qualification'.

The key requirements for any loop pre-qualification process are:

- obtaining an answer during the initial customer contact — this effectively means the pre-qualification process should last no more than 30 seconds,
- minimising the number of false decisions, so that service is not sold to a customer whose line cannot support it, and revenue is not lost by refusing service to a customer with a line that can support service,
- minimising the need for training both the customer service advisors³ (CSAs) and field engineers — this is achieved by avoiding the use of new test systems and equipment.

The use of an intelligent line termination device to meet these requirements has been mentioned in section 2.3. Other techniques for loop pre-qualification are also being investigated, including inferring line length, and hence insertion loss, from geographical location obtained from the customer's post code, and manual inspection of plant maps that show the routing and construction of the customer's line.

One of the more recent techniques to be investigated is the use of existing single-ended line test measurements on their own. The algorithm employed by this technique is straightforward.

³ Customer service advisor — this term is used to refer to the people who are the initial point of contact for customers calling with a sales enquiry or to report a problem with their service.

Firstly, the test results are checked to ensure no obvious faults exist on the line. Then, the average capacitance between each leg of the twisted pair and earth is found and a constant factor used to convert this capacitance to an insertion loss estimate.

The overall performance of this technique is illustrated in Figs 8a and 8b. In Fig 8a the format of the graph is shown. The ideal is for all points to lie within one of the 'good decision' quadrants. Figure 8b shows actual data obtained using the single-ended line test measurement technique. Error bars on the data necessitate a third category of classification, 'unknown'. Lines in this category need additional testing to determine if service can be provided. It can be seen from Fig 8b that, even with the 'unknown' region (defined by the error bars) set large enough to ensure no lines are in the 'incorrect decision' quadrants, there are still relatively few 'unknown lines'.

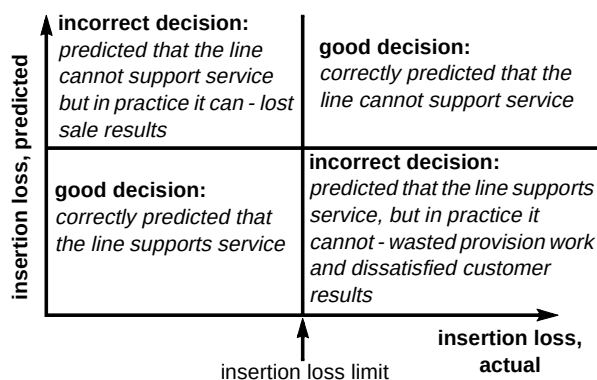


Fig 8a The four quadrants of the 'predicted loss' against 'actual loss' graph.

The overall performance of the line test system approach is better than can be obtained using the post code

information, and similar to that obtained from manual inspection of plant maps. It is also of significantly lower cost than either of these options as:

- the equipment is already installed and available to all customer service advisors, and it can measure the line of any customer who already has PSTN service,
- there are minimal training requirements (sales and field workforce) as the pre-qualification process can be 'hidden' from the customer service advisor — a simple 'yes/no/unknown' result can be presented.

A loop qualification process that exploits existing line test capabilities is shown in Fig 9.

This process offers similar or better overall performance than existing loop qualification process proposals and could be implemented at significantly lower cost.

3.2 Line test solutions for digital subscriber line technology

The ability to test the electrical condition of the twisted pair bearer is a fundamental requirement for effective management of DSL delivered services, as both fault localisation and network condition monitoring (as discussed in section 2) rely heavily on this information.

Using DSL can significantly complicate the measurement process, due, for example, to the use of splitter filters (ADSL) or the presence of unusual network termination equipment.

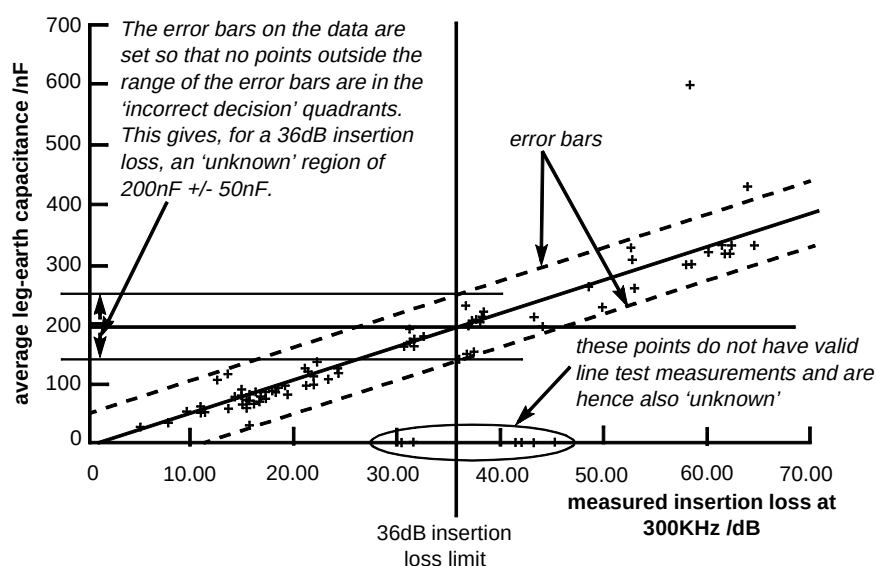


Fig 8b Typical performance of LTS capacitance data as a means of assessing insertion loss.

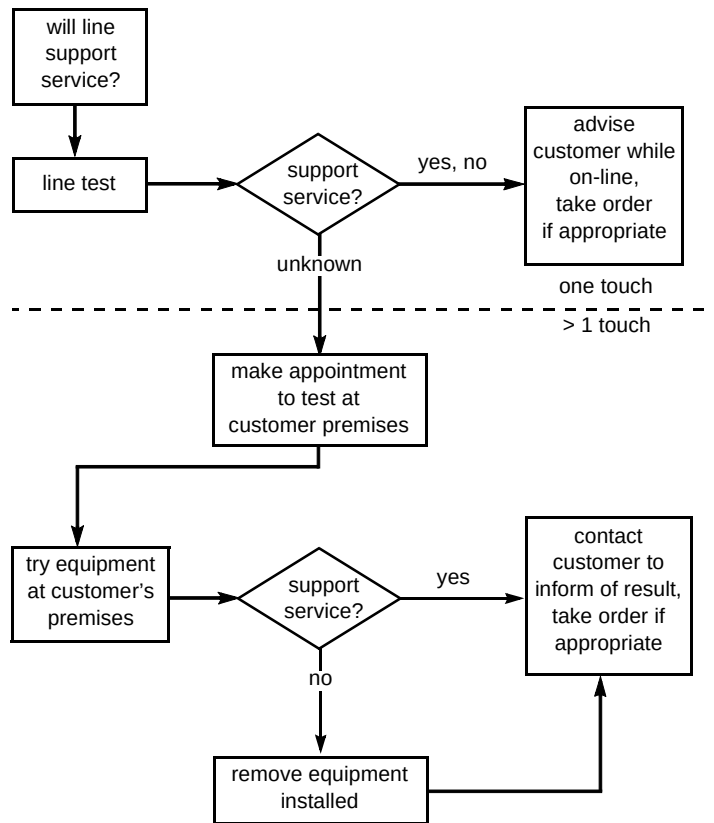


Fig 9 Proposal for a new loop pre-qualification process using only line test information and 'try-it-and-see' techniques.

The test solution for a recent new product within BT, 'HomeHighway', is now discussed to illustrate some of the problems and also novel methods of overcoming them.

HomeHighway uses ISDN transmission technology to provide a 144 kbit/s service similar to ISDN but targeted primarily at the residential market.

A key design feature of HomeHighway is that the existing PSTN service should continue to function, even after the installation of the network terminating equipment (NTE) at the customer premises, up until the line card is

changed from PSTN to ISDN at the local exchange (see Fig 10). This is achieved by building voltage detection into the NTE which senses when the line feed voltage increases from PSTN to ISDN levels. The NTE then automatically switches bypass relays so that service is routed through the ISDN transmission system.

The test solution being adopted for HomeHighway utilises the voltage detection feature to enable testing of the twisted-pair bearer to be performed using existing line test system capabilities. The exchange will disconnect the line from the ISDN line card using the normal test access relays.

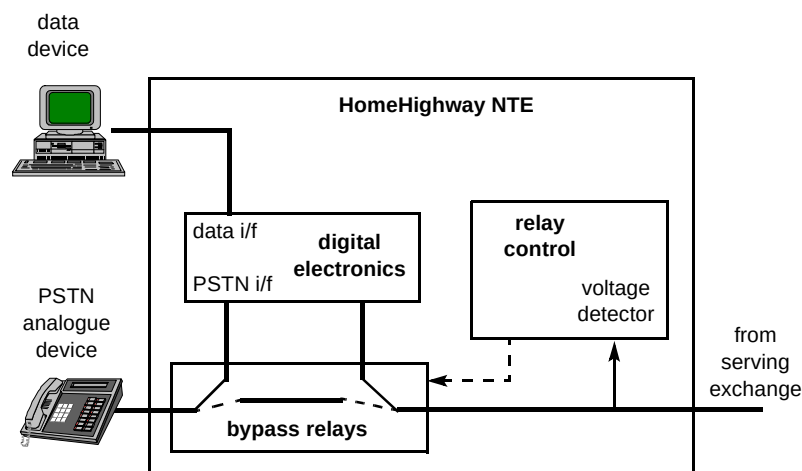


Fig 10 HomeHighway NTE design showing bypass relays.

The loss of ISDN line voltage will cause the NTE's voltage detector to activate the bypass relays, which will remain activated even when the standard PSTN line voltage is applied by the line test system. This gives a clean 'passive' metallic connection through to the customer's analogue devices. This innovative solution enables the same twisted pair bearer and customer equipment diagnostic algorithms to be used for HomeHighway as are already used for PSTN.

These algorithms, when combined with information obtained from ISDN test calls, provide a powerful diagnostic capability for the HomeHighway product.

4. New OAM technology for new networks

To minimise initial start-up costs, existing OAM solutions would ideally be applicable to a new network technology or service, as discussed in section 3. However, this constraint may mean missing opportunities offered by the new technology to reduce OAM costs or improve quality of service. Examples where new OAM solutions are delivering a significantly improved cost-to-quality-of-service ratio are now discussed.

4.1 Combining analogue and digital testing

Analogue PSTN delivery has no active components beyond the line card in the local exchange. For this reason, all diagnostic tests must be made by physically connecting measurement equipment to the line.

With DSL technology appearing in the access network, the situation changes dramatically. Active electronics are present at both ends of the network, and information about the condition of the line and customer's equipment can be obtained directly from the transmission system.

In practice, the optimum diagnostic situation is when the information from the transmission system can be combined with conventional tests of the bearer. The key challenge then becomes co-ordinating the two test activities and processing the information. The use of this approach is illustrated in Fig 11, which shows a typical ADSL testing scenario. The user interfaces with a test controller (perhaps via an intermediary customer service system) and requests a test. The test controller then initiates a digital test via the ADSL element manager. If necessary, it then instructs the ADSL element manager to switch the ADSL filters so that

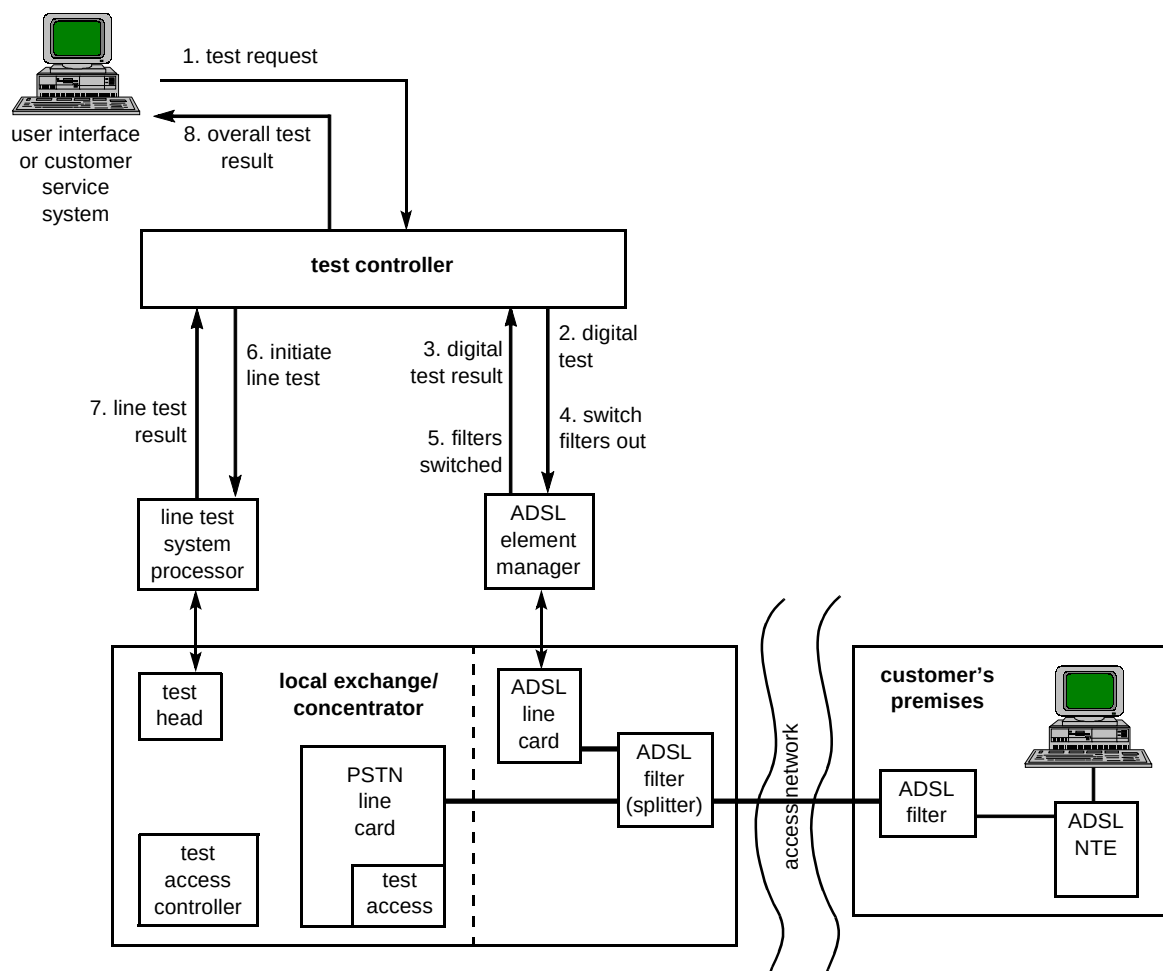


Fig 11 Using a test controller to co-ordinate multiple testing activities for ADSL.

they are out of circuit. This permits a conventional analogue line test to be performed using the existing line test and test access infrastructure. The test controller initiates this test, paying careful attention to handshaking and timing issues regarding the filter switching.

The test controller will collect together the test results, analyse them, and then present a simple diagnostic statement to the user.

Using a test controller not only ensures the diagnostic result obtained uses all available information, but also isolates the fault receptionist from the complexity of the underlying test process.

4.2 Testing Internet access

Traditionally, access network testing has consisted of testing the transmission bearer and passive customer apparatus. In section 4.1 the issue of testing at the data transmission level to provide additional diagnostic information was introduced.

In practice, customers are only interested in resolving a problem with their service — ‘Why doesn’t it work?’ — not with the underlying network. For PSTN the service is difficult to distinguish from the network and customer apparatus, but with more complex services over digital transmission systems, there is a clear difference. This is well illustrated by the example of Internet access services.

Figure 12 shows a typical architecture for Internet access. Customers using the Internet may experience a number of different problems while accessing the services they require and, in many cases, they will not be sure who to contact to obtain assistance. Often, the telecommunications company is the first organisation they call to help resolve

the problem. Figure 12 clearly shows, however, that the telecommunications company is only responsible for part of the network that is providing the service. Not only that, but the network is also, in general, very much more complex than that used to provide PSTN services. Worse still, routing through some parts of the network may not be known.

To offer high levels of support to customers using Internet-based applications, some form of Internet testing solution is desirable. This must be able to localise problems at least to the point where the appropriate customer support organisation has been identified. The Internet diagnostic tests that can be performed are generally well understood, but enabling these tests to be performed remotely by a customer service advisor with limited training, as is the case with basic PSTN testing, is a significant challenge.

It remains to be seen how Internet testing solutions will be incorporated into telcos’ networks. Ideally, the test functionality available will both enable telcos to offer high levels of customer service to Internet users and provide a competitive advantage. This is discussed in more detail in section 5.4.

5. What further opportunities exist?

The technologies discussed so far in this paper are either being introduced into the network or currently under development. This section discusses other opportunities for improving the OAM of new and existing networks that have yet to be investigated in detail.

5.1 Dealing with front-office call complexity

It is clear from many of the papers included in this edition of the Journal that the telecommunications industry

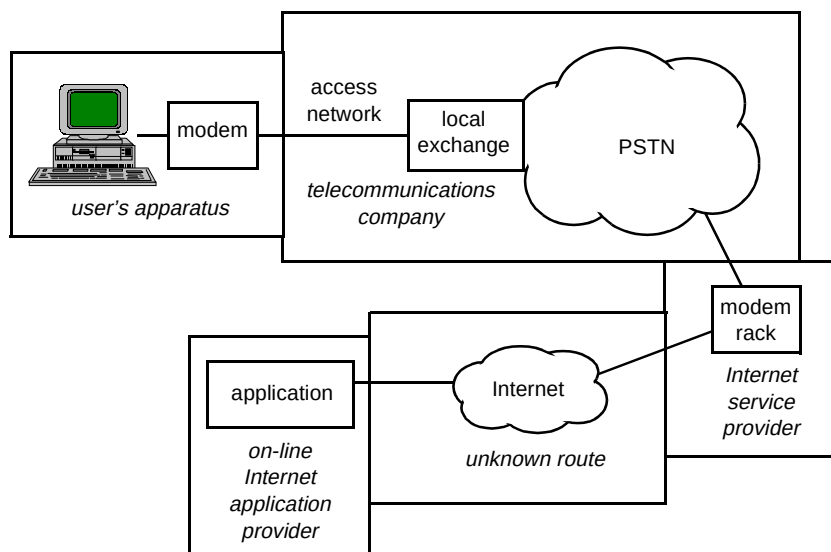


Fig 12 Typical architecture for accessing an on-line Internet application showing ownership of network components.

as a whole will experience a rapid increase in both the diversity and complexity of the technology and services in use.

It is essential that customer service advisors are able to support customers using these services at least as effectively as they are able to support the current dominant service, PSTN over twisted pairs.

It will be impractical to train all CSAs to diagnose problems with the full range of technologies in use. Both telcos and test solution suppliers are actively looking at the development of expert system technology to provide a decision support capability for CSAs.

An example of this, already being piloted within BT's business customer support organisation, utilises case-based reasoning technology to perform the analysis. The next generation of solutions, aimed primarily at supporting

residential customers, will most likely use additional techniques including neural networks and rule-based systems. This enables the most appropriate analysis technique to be used to solve various aspects of the diagnostic problem. The key issue is ensuring that the technologies are able to work together to form an overall conclusion.

5.2 Allocating work according to the skill required

Current methods of allocating fault reports to different skill groups of repair engineer often revolve around simple geographical boundaries, such as those shown in Fig 13a.

The introduction of computerised systems for the allocation of work to specific field engineers has allowed considerably more sophistication in managing resources and the skills they have. It is no longer realistic to think of only three skill groups, such as those shown in Fig 13a, making up the repair workforce.

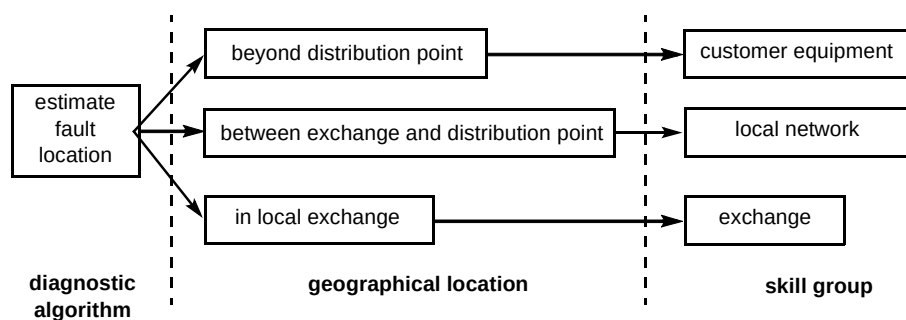


Fig 13a Using geographical location of fault to determine skill group required.

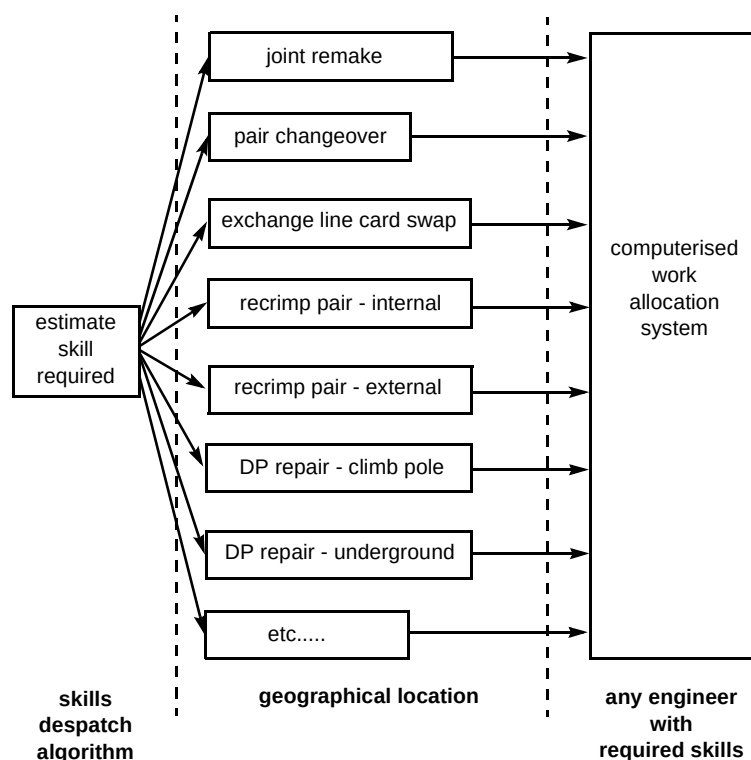


Fig 13b Using 'skills despatch algorithm' to allocate repair work.

New fault diagnosis algorithms are being researched that will estimate the skills required to restore service to a customer and, by passing this information into a computerised work allocation system, enable the job to be allocated to any engineer having the necessary skills. This is shown in Fig 13b. This approach will deliver significant improvements in resource utilisation and efficiency.

5.3 New line measurement techniques

The work discussed in this paper on twisted-pair access bearer testing has, in most cases, the same basic set of electrical measurements underpinning it. These are, primarily, measurements of resistance and capacitance between the two legs of the pair and between each leg and ground. These measurements are usually taken from the local exchange end of the pair.

It is likely that enhancing this set of measurements will give significant improvements in a telco's ability to localise network faults. One technique of particular interest is combining measurements of more than one pair. The main driver behind this is determining when two pairs are in contact with one another. This information, combined with knowledge of the routing of the pairs, can provide valuable location information as the fault must be in a portion of the network where the two pairs are routed in the same cable (see Fig 14).

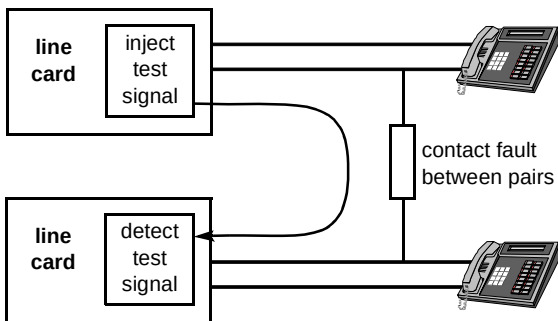


Fig 14 Localisation of contact faults using pair-to-pair contact detection.

The basic technique is to inject a known signal into a pair and monitor adjacent pairs in order to produce an estimate of the impedance between the pairs. A low impedance would suggest that the pairs are in contact with one another.

There are two significant technical hurdles to be overcome when using this technique:

- many exchanges only have one test access bus, so either signal injection or signal detection must be performed via the conventional speech path,

- some prior knowledge is required about which lines may be connected to other lines, as testing the impedance between all combinations of two lines would take a prohibitively long time.

Current development of this technique is concentrating on identifying solutions to these two issues.

5.4 Positioning test capability as a premium product

Traditionally, network test capability has existed to improve a telecommunications company's ability to deliver high quality of service levels at an acceptable cost. Although the test capability is usually invisible to the customer, their quality of service expectations are now such that it forms an essential part of the product they purchase.

With new services becoming available to customers, an opportunity exists to offer different levels of service with increased levels of test capability, and hence quality of service and support levels, at a premium price. A good illustration of this is in the area of Internet testing. A possible scenario is illustrated in Fig 15. The base level product is dial-up access using an analogue modem. The telco's obligation is the same as that for PSTN — namely, to ensure that the condition of the bearer is acceptable for voice transmission. Customers who wish to have a guaranteed level of data service can move to ISDN-based access. Not only are higher data rates available, but the telco will test both the bearer and the data transmission system.

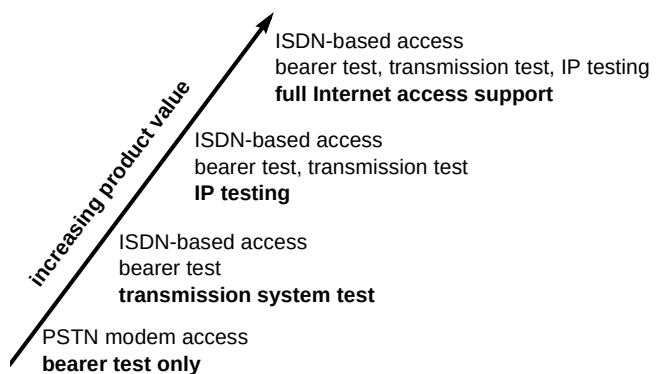


Fig 15 Using test capability to enhance product value.

A next level product could include IP testing capabilities (as discussed in section 4.2). With this service, the telco would not only ensure the data transmission was working correctly but would also help diagnose problems at the IP level.

The highest-level product would be a full Internet access support service, which would aim to resolve any issues associated with accessing Internet services, including diagnosing problems with the user's own system set-up.

It is likely that using test capability to enhance basic product offerings in this way, and hence increase revenue potential, will be essential if telcos are to provide advanced test solutions for the more complex service offerings. This is because building advanced test capabilities into the cost of the base product could make it uncompetitive.

6. Conclusions

A wide range of innovative research and development work is being undertaken to ensure that telcos are able to operate both existing and new technology cost-effectively. Much of the work described is already showing significant returns on the research and development investments made.

It is essential that an appropriate balance of investment is struck between developing new network technology and services, and developing solutions for ensuring high levels of operational efficiency.

In addition, it is also important to apportion the investment appropriately across the three themes described in Fig 1. In particular, opportunities for using new technology to improve the operational efficiency of existing networks and services must not be overlooked.

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